K-Nearest Neighbor Classification

Predicting Stroke Risk in Hospital Patients

Research Question

Is it possible to predict which hospital patients are at high risk of stroke in order to warn them and take steps to avoid?

Analysis Goal

This data analysis aims to build a KNN model able to determine whether a patient will be considered at high risk to experience a stroke.

Method Justification

KNN functions by looking at however many nearby data points are set as the K value, and the predicted data point will be whichever value the majority of those nearby data points consist of. The expected outcome will become the weighted average of the known outcome of those nearby data points, based on similarity.

KNN assumes that the data exists in feature space, and can therefore be measured using distance metrics.

Packages & Libraries

- **Pandas** , This allows transformation of the dataset from a csv into a dataframe that can be manipulated and analyzed.
- **Numpy** , Numpy will allow parameters to be set using the arange function.
- **sklearn.model_selection train_test_split** , This package is necessary in order to create and analyze a testing data set.
- **sklearn.preprocessing StandardScaler** , This will allow for scaling of data.
- **sklearn.neighbors KNeighborsClassifier** , This includes the KNN function used to build our model.
- **sklearn.metrics confusion_matrix** , Allows us to build a confusion matrix from selected data.
- **sklearn.model_selection GridSearchCV** , This will allow for parameter selection for the KNN model.
- **sklearn.pipeline Pipeline** , This will allow sequential transformations of data in preprocessing.
- **sklearn.metrics roc_auc_score** , This allows for area under the curve calculations.
- **Seaborn** , This is used to graph data distribution for outlier analysis.

Data Preprocessing

The dataset categories must be converted to dummy variables and one hot encoded.

Variables

Numeric: Age, Population, Vitamin D levels

Categorical: Gender, Area, Marital, ReAdmis, Initial_admin, Complication_risk, Services, Arthritis, Hyperlipidemia, Asthma, Anxiety, Stroke, Soft_drink, Diabetes, Overweight, HighBlood

Data Preparation Steps

Line 10: Duplicates will be evaluated using .duplicate function, but no fields populate that are necessary to change.

Lines 12-32: All categorical columns are evaluated using .unique to ensure there are no duplicates, nulls, or values in need of conversion, and these all look good to proceed.

Line 35: DropNa function run over entire dataset to ensure no invalid values.

Lines 37-71: Outlier analysis is also performed on all numeric columns to be used in the initial logistic model, and outliers are replaced with the column mean.

Lines 80-82: The data has many True/False columns. Data analysis will be much easier if these are converted to 1/0. This is achieved by running a .replace function over the entire dataframe converting True/False to 1/0. This is done after creating a dummies dataframe to ensure to the dummy values are one hot encoded.

Lines 84-87: Redundant dummy columns are dropped and remaining columns are appropriately named.

Lines 90-95: Lastly, the numeric data columns will be scaled before splitting for training and testing.

Analysis

K nearest neighbors has been used to analyze this data set by using nearby variables as predictors for the predicted data point, 'Stroke'. The data points from patients who have experienced strokes are now able to provide a roughly 80% accurate estimation of whether patients in the known data set will have experienced a stroke or not. This can now provide statistically valid recommendations to help predict and save future stroke patients.

of whether patients in the known data set will have experienced a stroke or not. This ca
now provide statistically valid recommendations to help predict and save future stroke patients.

```
In [48]:
    ...:
    ...: print("Training Accuracy: ", knncv.score(X_train, y_train))
    ...: print("Testing Accuracy: ", knncv.score(X_test, y_test))
Training Accuracy:  0.800625
Testing Accuracy:  0.8005

In [49]:
    ...:
    ...: knn = KNeighborsClassifier(n_neighbors = 12)
    ...: knn.fit(X_train, y_train)
    ...: print("KNN Model Accuracy: ")
    ...: print(knn.score(X_test, y_test))
    ...: y_predicted = knn.predict(X_test)
    ...: print("Confusion Matrix: ")
    ...: print(confusion_matrix(y_test, y_predicted))
    ...: y_predicted_probability = knn.predict_proba(X_test)[:,1]
    ...: print("KNN model AUC: ")
    ...: print(roc_auc_score(y_test, y_predicted_probability))
KNN Model Accuracy:
0.7995
Confusion Matrix:
[[1596    5]
 [ 396    3]]
KNN model AUC:
0.48397070126910025

In [50]:
```

KNN model accuracy, confusion matrix, AUC, and GridSearchCV output

Accuracy & AUC

The accuracy score of 0.7995 means that 79.95% of the predictions made by the model are correct.

The AUC score of 0.483 means that the model successfully distinguishes between classes at a 48.3% rate.

Results & Implications

The model accuracy score of 80% is very promising. This model could certainly be used to successfully alert many patients that they are at higher risk of stroke, and should be careful and prepared for an emergency.

The AUC score of 0.483 implies that the ROC is successfully distinguishing between positive or negative classes a bit less than half the time, so the model is unable to provide relevant distinction between true positives and false positives. The AUC is derived from the ROC curve, plotting true positives and false positives against one another, showing that the model accurately predicts true positives 48.3% of the time. This is not problematic, as false positives will not hurt patients, and alerting those patients who will experience a stroke is potentially life saving.

Many of the model features are things patients do have at least some control over, such as soft drink consumption, area population, and overweight status, if patients are at high risk of a stroke, it may be worth seriously considering some life changes that could push them out of the higher risk population.

Many of the features could be strong long term indicators, allowing doctors to recommend courses of action to reduce stroke risk.

Limitations

This data analysis could be limited by the smaller sample size of the testing data. Using only 2000 of the 10,000 records could result in statistical biases due to the sample size not being large enough to accurately convey summary statistics.

Course of Action

Based on this model data, it would be prudent to alert patients at high risk of stroke, as the model is accurately predicting 80% of known outcomes. Patients can hopefully take steps to avoid or prepare for the stroke and minimize the risk and damage.

The AUC score being 0.483 shows that the model cannot distinguish well between classes, but that is okay for the scope of this project. There will be many false positives, people who are at higher risk of stroke, who never experience a stroke. As there is no serious downside to a false positive in this scenario, the ability to filter out false positives is not nearly as important as alerting the patients who will suffer a stroke.

If desired, this model could likely be improved by testing out independent variables and assessing their importance. The most relevant features would be available to the hospital with a few more steps of analysis, and recommendations for patient lifestyle could potentially be made based off of those.

Tools: Python, pandas, numpy, seaborn, sklearn